THE SHRIEK OF THE STRUCTURE SHEAF ALONG A FLAT MORPHISM IS RELATIVELY PERFECT AND SELF-DUAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0E9T (sources/stacks-duality/pinned-original-source.tex line 4890).

Allowed standard prerequisites (editorial).

- The local calculation of the shriek on affines. Same chapter, Remark labelled remark-local-calculation-shriek, cited at proof lines 4901-4904. Turns the statement into an algebraic one.
- The algebraic statement for a flat ring map. Dualizing Complexes, Lemma labelled dualizing-lemma-relative-dualizing-algebraic, cited at proof lines 4904-4906. Supplies both conclusions in the affine case.
- Comparison of relatively perfect complexes with the algebraic notion. Derived Categories of Schemes, Lemma labelled perfect-lemma-affine-locally-rel-perfect, cited at proof lines 4904-4906. Matches relative perfection with the algebraic hypothesis.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0BV2: sources/stacks-duality/pinned-original-source.tex, source lines 4636–4636. Same-chapter pointer into the author environment remark at source line 4636; invoked at proof line 4901 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BZW: sources/stacks-chow/tags.txt, source lines 15066–15066. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4903 of the counted proof. It is

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Source reference. Original lemma 0DI2: sources/stacks-chow/tags.txt, source lines 16927–16927. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4905 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. If f is flat, then $f^!\mathcal{O}_Y$ is a Y-perfect object of $D(\mathcal{O}_X)$ and $\mathcal{O}_X \to R\mathcal{H}om_{\mathcal{O}_X}(f^!\mathcal{O}_Y, f^!\mathcal{O}_Y)$ is an isomorphism.*

Proof. Both assertions are local on X. Thus we may assume X and Y are affine. Then Remark 0BV2 turns the lemma into an algebra lemma, namely Dualizing Complexes, Lemma 0BZW. (Use Derived Categories of Schemes, Lemma 0DI2 to match the languages.) □